INCH-POUND

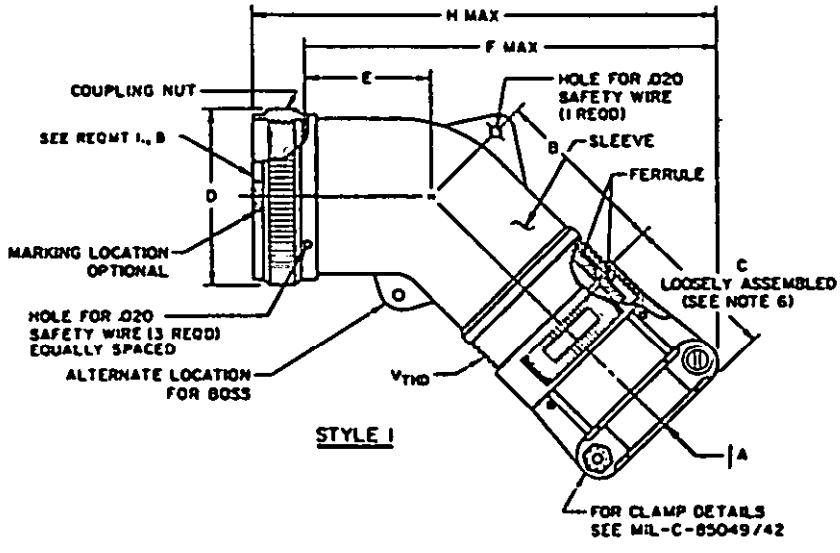
MIL-C-85049/23A
30 November 1990
SUPERSEDING
MIL-C-85049/23
16 January 1980
MS3189B
10 May 1982

MILITARY SPECIFICATION SHEET

CONNECTOR ACCESSORIES, ELECTRICAL, BACKSHELL, NONENVIRONMENTAL, 45°, SHIELD TERMINATION, CATEGORY 3A (FOR MIL-C-5015 CRIMP, MIL-C-26482 SERIES 2, MIL-C-81703 SERIES 3, AND MIL-C-83723 SERIES III CONNECTORS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-85049.



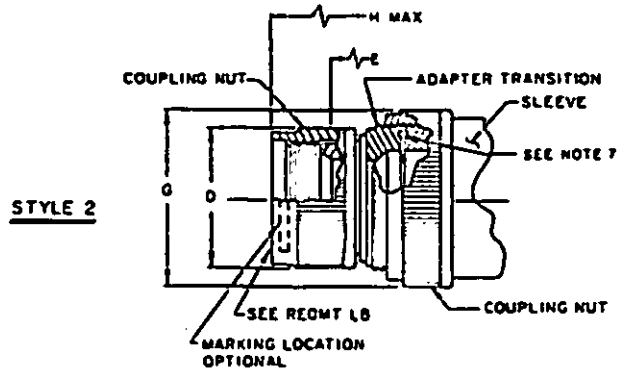


FIGURE 1. Configuration and dimensions.

AMSC N/A
FSC 5935

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TABLE I. Dimensions.

Dash No.	Fits Shell of Connector				Clamp Size	A (see note 6) Wire Bundle Accommodation Range		Style	B ±.125	C Ref	D Dia Max
	MIL-C-81703 Ser 3	MIL-C-26482 Ser 2	MIL-C-5015 Crimp	MIL-C-83723 Ser III		Min	Max				
1	3				4	.125	.250	1	1.790	1.027	.669
2					6	.250	.437	2	1.910	1.027	.669
3		8	8S	8	4	.125	.250	1	1.182	1.027	.617
4		10	10S, 10SL	10	4	.125	.312	1	1.850	1.027	.734
5					6	.250	.375	1	1.245	1.027	.734
6	7	12	12,12S	12	4	.125	.312	1	1.910	1.027	.858
7					6	.250	.437	1	1.910	1.027	.858
8					10	.350	.500	1	1.910	1.027	.858
9	12	14	14,14S	14	6	.250	.437	1	2.000	1.027	.984
10					10	.350	.575	1	1.655	1.027	.984

Dash No.	E ±.125	V Thread Class 2A	F Max	G Dia Max	H Max
1	.561	.625-24 UNEF	3.00	-	3.39
2	1.375	.750-20 UNEF	3.97	.858	4.36
3	.500	.625-24 UNEF	2.52	-	2.91
4	.561	.625-24 UNEF	3.05	-	3.44
5	.561	.750-20 UNEF	2.69	-	3.08
6	.625	.625-24 UNEF	3.16	-	3.55
7	.625	.750-20 UNEF	3.22	-	3.61
8	.625	1.000-20 UNEF	3.29	-	3.68
9	.688	.750-20 UNEF	3.35	-	3.74
10	.688	1.000-20 UNEF	3.17	-	3.56

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TABLE I. Dimensions (continued).

Dash No.	Fits Shell of Connector				Clamp Size	A (see note 6) Wire Bundle Accommodation Range		Style	B ±.125	C Ref	D Dia Max
	MIL-C-81703 Ser 3	MIL-C-26482 Ser 2	MIL-C-5015 Crimp	MIL-C-83723 Ser III		Min	Max				
11	19	16	16,16S	16	6	.250	.437	1	2.110	1.027	1.112
12					12	.500	.700	1	1.750	1.059	1.112
13	27	18	18	18	10	.350	.625	1	2.300	1.027	1.218
14					16	.625	.779	1	1.938	1.156	1.218
15	37	20	20	20	10	.350	.625	1	2.300	1.027	1.345
16					16	.625	.904	1	1.938	1.156	1.345
17		22	22	22	12	.500	.750	1	2.450	1.059	1.468
18					20	.875	1.029	1	2.093	1.375	1.468
19		24	24	24	12	.500	.750	1	2.450	1.069	1.593
20					20	.875	1.144	1	2.093	1.375	1.593

Dash No.	E ±.125	V Thread Class 2A	F Max	G Dia Max	H Max
11	.781	.750-20 UNEF	3.52	-	3.91
12	.781	1.1875-18 UNEF	3.43	-	3.82
13	.968	1.000-20 UNEF	3.91	-	4.30
14	.968	1.4375-18 UNEF	3.90	-	4.29
15	.968	1.000-20 UNEF	3.91	-	4.30
16	.968	1.4375-18 UNEF	3.90	-	4.29
17	1.063	1.1875-18 UNEF	4.21	-	4.60
18	1.063	1.750-10 UNS	4.38	-	4.77
19	1.063	1.1875-18 UNEF	4.21	-	4.60
20	1.063	1.750-18 UNS	4.38	-	4.77

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TABLE I. Dimensions (continued).

Dash No.	Fits Shell of Connector: MIL-C-81703 Ser 3	MIL-C-26482 Ser 2	MIL-C-5015 Crimp	MIL-C-83723 Ser III	Clamp Size	A (see note 6) Wire Bundle Accommodation Range Min	A (see note 6) Wire Bundle Accommodation Range Max	Style	B ±.125	C Ref	D Dia Max
21			28		16	.625	.937	1	2.500	1.156	1.969
22					24	1.000	1.375	1	2.156	1.500	1.969
23			32		16	.625	.937	1	2.600	1.156	2.219
24					20	.875	1.250	1	2.600	1.375	2.219
25					28	1.250	1.625	1	2.250	1.781	2.219
26			36		16	.625	.937	1	2.690	1.156	2.469
27					24	1.000	1.375	1	2.690	1.500	2.469
28					32	1.437	1.840	1	2.344	1.830	2.489
29			40		16	.625	.937	1	2.690	1.156	2.719
30					24	1.000	1.375	1	2.690	1.500	2.719

Dash No.	E ±.125	V Thread Class 2A	F Max	G Dia Max	H Max
21	1.188	1.4375-18 UNEF	4.52	-	5.05
22	1.188	2.000-18 UNS	4.72	-	5.23
23	1.250	1.4375-18 UNEF	4.65	-	5.18
24	1.250	1.750-18 UNS	4.92	-	5.45
25	1.250	2.250-16 UN	5.19	-	5.72
26	1.281	1.4375-18 UNEF	4.74	-	5.27
27	1.281	2.000-18 UNS	5.19	-	5.72
28	1.281	2.500-18 UN	5.41	-	5.94
29	2.031	1.4375-18 UNEF	5.49	-	6.02
30	2.031	2.000-18 UNS	5.94	-	6.47

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TABLE I. Dimensions (continued).

Dash No.	Fits Shell of Connector				Clamp Size	A (see note 6) Wire Bundle Accommodation Range		Style	B ±.125	C Ref	D Dia Max
	MIL-C-81703 Ser 3	MIL-C-26482 Ser 2	MIL-C-5015 Crimp	MIL-C-83723 Ser III		Min	Max				
31			40		32	1.437	1.875	1	2.344	1.830	2.719
32			44		16	.625	.937	1	2.690	1.156	2.969
33					24	1.000	1.375	1	2.690	1.500	2.969
34					32	1.437	1.875	1	2.344	1.830	2.969
35			48		16	.625	.937	1	2.690	1.156	3.219
36					24	1.000	1.375	1	2.690	1.500	3.219
37					32	1.437	1.875	1	2.344	1.830	3.219
38	61				12	.500	.750	1	2.450	1.059	1.653
39					20	.875	1.184	1	2.093	1.375	1.653
40	19	16	16,16S	16	10	.350	.625	1	2.110	1.027	1.112

Dash No.	E ±.125	V Thread Class 2A	F Max	G Dia Max	H Max
31	2.031	2.500-16 UN	6.16	-	6.69
32	2.031	1.437-18 UNEF	5.49	-	6.02
33	2.031	2.000-18 UNS	5.94	-	6.47
34	2.031	2.500-16 UN	6.16	-	6.69
35	2.031	1.4375-18 UNEF	5.49	-	6.02
36	2.031	2.000-18 UNS	5.94	-	6.47
37	2.031	2.500-16 UN	6.16	-	6.69
38	1.063	1.1875-18 UNEF	4.21	-	4.60
39	1.063	1.750-18 UNS	4.38	-	4.77
40	.781	1.000-20 UNEF	3.59	-	3.98

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TABLE I. Dimensions (continued).

Dash No.	Fits Shell of Connector				Clamp Size	A (see note 6) Wire Bundle Accommodation Range		Style	B ±.125	C Ref	D Dia Max
	MIL-C-81703 Ser 3	MIL-C-26482 Ser 2	MIL-C-5015 Crimp	MIL-C-83723 Ser III		Min	Max				
41	27	18	18	18	4	.125	.312	1	2.300	1.027	1.218
42					6	.250	.437	1	2.300	1.027	1.218
43	37	20	20	20	6	.250	.437	1	2.300	1.027	1.345
44		22	22	22	4	.125	.312	1	2.450	1.027	1.488
45					6	.250	.437	1	2.450	1.027	1.468
46		24	24	24	10	.350	.625	1	2.450	1.027	1.593
47			36		12	.500	.750	1	2.690	1.059	2.469
48			40		12	.500	.750	1	2.690	1.059	2.719
49		10	10S, 10SL	10	6	.250	.437	2	1.910	1.027	.734
50	12	14	14,14S	14	12	.500	.750	2	2.300	1.059	.984

Dash No.	E ±.125	V Thread Class 2A	F Max	G Dia Max	H Max
41	.968	.625-24 UNEF	3.80	-	4.19
42	.968	.750-20 UNEF	3.84	-	4.23
43	.968	.750-20 UNEF	3.84	-	4.23
44	1.063	.625-24 UNEF	3.98	-	4.37
45	1.063	.750-20 UNEF	4.04	-	4.43
46	1.063	1.000-20 UNEF	4.11	-	4.50
47	1.281	1.1875-18 UNEF	4.60	-	5.13
48	2.031	1.1875-18 UNEF	5.35	-	5.88
49	1.375	.750-20 UNEF	3.97	.858	4.36
50	1.717	1.1875-18 UNEF	4.76	1.218	5.15

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TABLE I. Dimensions (continued).

Dash No.	Fits Shell of Connector				Clamp Size	A (see note 6) Wire Bundle Accommodation Range		Style	B ±.125	C Ref	D Dia Max
	MIL-C-81703 Ser 3	MIL-C-26482 Ser 2	MIL-C-5015 Crimp	MIL-C-83723 Ser III		Min	Max				
51	19	16	16,16S	16	16	.625	.937	2	2.450	1.156	1.112
52	27	18	18	18	12	.500	.750	1	2.300	1.059	1.218
53	61				16	.625	.937	1	2.450	1.156	1.653
54	37	20	20	20	12	.500	.750	1	2.300	1.059	1.345
55					20	.875	1.250	2	2.500	1.375	1.345
56		22	22	22	10	.350	.625	1	2.450	1.027	1.468
57					16	.625	.937	1	2.450	1.156	1.468
58					20	.875	1.250	2	2.500	1.375	1.468
59		24	24	24	16	.625	.937	1	2.450	1.156	1.593
60			28		12	.500	.750	1	2.500	1.059	1.969

Dash No.	E ±.125	V Thread Class 2A	F Max	G Dia Max	H Max
51	1.812	1.4375-18 UNEF	5.11	1.468	5.50
52	.968	1.1875-18 UNEF	4.01	-	4.40
53	1.063	1.4375-18 UNEF	4.36	-	4.75
54	.968	1.1875-18 UNEF	4.01	-	4.40
55	1.936	1.750-18 UNS	5.54	1.969	5.93
56	1.063	1.000-20 UNEF	4.11	-	4.50
57	1.063	1.4375-18 UNEF	4.36	-	4.75
58	1.936	1.750-18 UNS	5.54	1.969	5.93
59	1.063	1.4375-18 UNEF	4.36	--	4.75
60	1.188	1.1875-18 UNEF	4.37	--	4.90

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TABLE I. Dimensions (continued).

Dash No.	Fits Shell of Connector: MIL-C-81703 Ser 3	MIL-C-26482 Ser 2	MIL-C-5015 Crimp	MIL-C-83723 Ser III	Clamp Size	A (see note 6) Wire Bundle Accommodation Range Min	A (see note 6) Wire Bundle Accommodation Range Max	Style	B ±.125	C Ref	D Dia Max
61			28		20	.875	1.250	1	2.500	1.375	1.969
62			32		24	1.000	1.375	1	2.600	1.500	2.219
63			36		20	.875	1.250	1	2.690	1.375	2.469
64					28	1.250	1.625	1	2.690	1.781	2.469
65			40		20	.875	1.250	1	2.690	1.375	2.719
66					28	1.250	1.625	1	2.690	1.781	2.719
67			44		20	.875	1.250	1	2.690	1.375	2.969
68					28	1.250	1.625	1	2.690	1.781	2.969
69			48		20	.875	1.250	1	2.690	1.375	3.219
70					28	1.250	1.625	1	2.690	1.781	3.219

Dash No.	E ±.125	V Thread Class 2A	F Max	G Dia Max	H Max
61	1.188	1.750-18 UNS	4.79	-	5.32
62	1.250	2.000-18 UNS	5.10	-	5.63
63	1.281	1.750-18 UNS	5.02	-	5.55
64	1.281	2.250-16 UN	5.54	-	6.07
65	2.031	1.750-18 UNS	5.77	-	6.30
66	2.031	2.250-16 UN	6.29	-	6.82
67	2.031	1.750-18 UNS	5.77	-	6.30
68	2.031	2.250-16 UN	6.29	-	6.82
69	2.031	1.750-18 UNS	5.77	-	6.30
70	2.031	2.250-16 UN	6.29	-	6.82

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TABLE I. Dimensions (continued).

Dash No.	Fits Shell of Connector				Clamp Size	A (see note 6) Wire Bundle Accommodation Range		Style	B ±.125	C Ref	D Dia Max
	MIL-C-81703 Ser 3	MIL-C-26482 Ser 2	MIL-C-5015 Crimp	MIL-C-83723 Ser III		Min	Max				
71	7	12	12,12S	12	10	.350	.625	2	2.110	1.027	.858
72	27	18	18	18	16	.625	.937	2	2.450	1.156	1.218
73		24	24	24	20	.875	1.250	2	2.500	1.375	1.593
74	12	14	14,14S	14	4	.125	.312	1	2.000	1.027	.984
75	19	16	16,16S	16	4	.125	.312	1	2.110	1.027	1.112

Dash No.	E ±.125	V Thread Class 2A	F Max	G Dia Max	H Max
71	1.529	1.000-20 UNEF	4.33	1.112	4.72
72	1.812	1.4375-18 UNEF	5.10	1.468	5.49
73	1.936	1.750-18 UNS	5.53	1.969	5.92
74	.688	.625-24 UNEF	3.28	-	3.67
75	.781	.625-24 UNEF	3.45	-	3.84

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INCH	mm	INCH	mm	INCH	mm	INCH	mm
0.005	0.13	1.245	31.62	2.600	66.04	4.720	119.88
0.020	0.51	1.250	31.75	2.690	68.33	4.750	120.65
0.062	1.57	1.281	32.54	2.719	69.06	4.760	120.90
0.250	6.35	1.345	34.16	2.969	75.41	4.770	121.15
0.312	7.92	1.375	34.93	3.000	76.20	4.790	121.67
0.350	8.89	1.437	36.50	3.160	80.26	4.900	124.46
0.375	9.53	1.4375	36.51	3.170	80.52	4.920	124.97
0.437	11.10	1.465	37.21	3.219	81.76	5.050	128.27
0.500	12.70	1.468	37.29	3.220	81.79	5.110	129.79
0.561	14.25	1.500	38.10	3.280	83.31	5.130	130.30
0.575	14.60	1.529	38.84	3.290	83.57	5.150	130.81
0.617	15.67	1.593	40.46	3.350	85.09	5.180	131.57
0.625	15.88	1.625	41.28	3.430	87.12	5.190	131.83
0.669	16.99	1.653	41.99	3.450	87.63	5.230	132.84
0.688	17.48	1.655	42.04	3.590	91.19	5.270	133.85
0.700	17.78	1.717	43.61	3.670	93.22	5.370	135.12
0.734	18.64	1.750	44.45	3.800	96.52	5.350	135.89
0.750	19.05	1.781	45.24	3.820	97.03	5.410	137.41
0.779	19.79	1.790	45.47	3.840	97.54	5.450	138.43
0.781	19.84	1.812	46.02	3.900	99.06	5.490	139.45
0.858	21.79	1.830	46.48	3.910	99.31	5.500	139.75
0.875	22.22	1.840	46.74	3.970	100.84	5.530	140.46
0.904	22.96	1.850	46.99	3.980	101.09	5.540	140.72
0.937	23.80	1.910	48.51	4.010	101.85	5.550	140.97
0.968	24.59	1.936	49.17	4.040	102.62	5.630	143.00
0.984	24.99	1.938	49.23	4.110	104.39	5.720	145.28
1.000	25.40	1.969	50.01	4.190	106.42	5.770	146.55
1.027	26.09	2.000	50.80	4.210	106.93	5.880	149.35
1.029	26.14	2.031	51.59	4.230	107.44	5.920	150.36
1.059	26.90	2.093	53.16	4.290	108.96	5.930	150.62
1.063	27.00	2.110	53.59	4.300	109.22	5.940	150.88
1.112	28.24	2.156	54.76	4.360	110.74	6.020	152.90
1.144	29.06	2.219	56.36	4.370	110.99	6.070	154.17
1.156	29.36	2.250	57.15	4.380	111.25	6.160	156.46
1.182	30.02	2.300	58.42	4.400	111.76	6.290	159.77
1.184	30.07	2.344	59.54	4.430	112.52	6.300	160.02
1.1875	30.16	2.450	62.23	4.500	114.30	6.470	164.33
1.188	30.18	2.469	62.71	4.520	114.81	6.660	169.16
1.200	30.48	2.500	63.50	4.600	116.84	6.690	169.92
1.218	30.94	2.520	64.01	4.650	118.11	6.820	173.22

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REQUIREMENTS:

1. Design and construction:

   a. Dimensions and configuration: See figure 1 and table I.

   b. Interface dimensions shall conform to MIL-C-85049 figure 4.

   c. Figure 1 consists of a coupling nut and follower. The coupling nut shall be captivated to and free to rotate on the follower.

2. Material and finish:

   a. Accessory components shall be as shown in table II.

   b. Clamp assembly screws - Corrosion resistant steel in accordance with MIL-C-85049, passivated per QQ-P-35.

3. Qualification: See MIL-C-85049, category 3A. Information pertaining to qualification of products may be obtained from the Commanding Officer, Naval Avionics Center, Attention: Code B/444, 6000 East 21st Street, Indianapolis, IN 46219-2189.

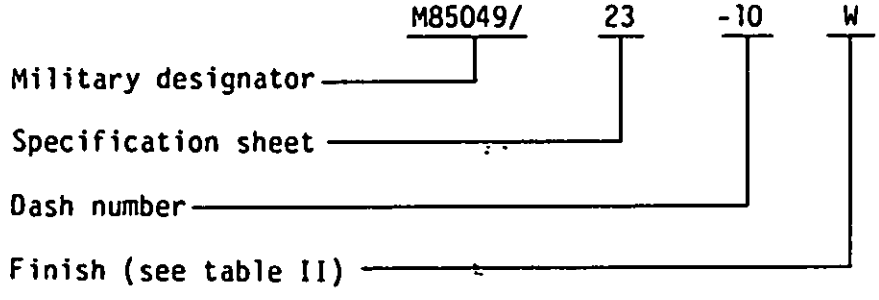
4. Applications:

   a. These accessories are for shield termination.

   b. Style 2 applies when the cable diameter exceeds the "E" diameter of the interface dimensions defined in MIL-C-85049 figure 4.

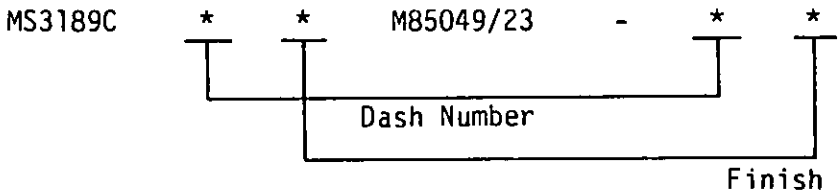
   c. For assembly instructions see figure 2 of MIL-C-85049/25.

5. Part Number: See example below:



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6. Supersession data:



Old	New
A	W
C	W

TABLE II. Material and finish.

Component	Material	Finish
Body	Aluminum alloy in accordance with MIL-C-85049	1/ N, 2/ W
Body	Corrosion resistant steel in accordance with MIL-C-85049	B, S

1/ Not for Navy use. Restricted to Air Force Space applications only.
2/ W finish is not for use in space application.

NOTES:

1. Dimensions are in inches.
2. Metric equivalents shall be based upon 1.00 inch = 25.4 mm and shall be in accordance with ASTM E380, paragraph 4.5, Method A and ASTM E29, rounding-off method.
3. Unless otherwise specified, tolerances are ±.005 for three place decimals and ±.02 for two place decimals.
4. Dimensions apply after plating.
5. Loosely assembled is defined as the MIL-C-85049/42 clamp tightened to the MIL-C-85049/23 connector accessory body a minimum of two full thread lengths.
6. "Wire bundle accommodation range" dimension is defined as the envelope area for the wire bundle. This dimension is not meant to define the clamp hardware limits.
7. O-ring is not supplied for style 2.

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Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities:
Army - AR, MI
Navy - EC
Air Force - 11, 99
DLA - ES

Preparing activity:
Navy - AS
(Project No. 5935-3729-09)

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